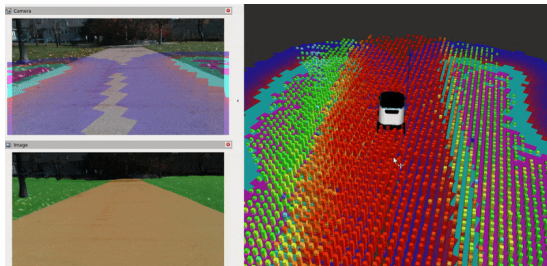


Perception: sensing lanes and obstacles

- **LiDAR** (Velodyne VLP-16, 10 Hz) — physical obstacles in 3-D; any color, any lighting
- **Vision** (ZED X) — three color classes: white lanes, barrels, and white potholes
- White potholes *separated* from white lane lines — one avoided, one followed
- Startup auto-calibration adapts to the day's lighting; fused into one Nav2 costmap

Requirement (IGVC §II.2): detect obstacles ≥ 2 m $\rightarrow$ **15 m**; vision ≥ 10 Hz $\rightarrow$ **10 Hz**.



live lane / barrel / pothole segmentation $\rightarrow$ point-cloud costmap



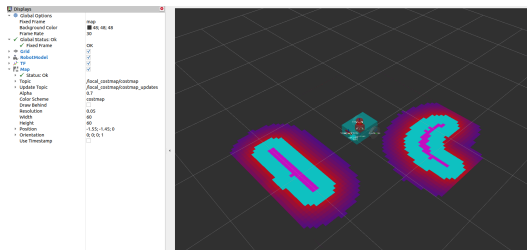
VLP-16 LiDAR



ZED X camera

Driving logic: plan, then re-plan continuously

- **Planner** — Navfn (Dijkstra): chosen over Smac-Hybrid, whose 2.31 m turning circle can't fit a 10-ft lane; our tracked chassis turns in place
- **Controller** — MPPI *generates* trajectories at 20 Hz to deviate around new barrels, not just follow the plan
- Handles potholes, switchbacks, dead-ends and the ramp; recovery tree under a 45 s watchdog (60 s hold-up rule)
- Dual EKF pose; mission_manager fires GPS waypoints in sequence



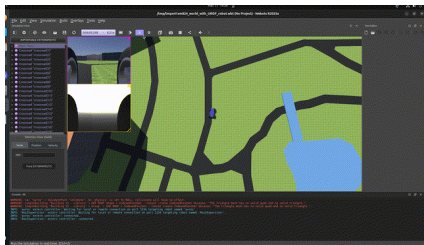
Nav2 local costmap: lethal cells (cyan) ringed by the inflation cost-gradient

Requirement obstacle reaction ≤ 250 ms $\rightarrow$ **47 ms**;

waypoint accuracy ≤ 1 m; avg speed ≥ 1 mph $\rightarrow$ **1.7 mph**.

Validation: key performance indicators

Two-tier testing — a **Webots** simulation validates the full Nav2 pipeline; the **hardware** chassis proves the numbers. KPIs are set against the IGVC rules.



Webots: full-pipeline campus simulation

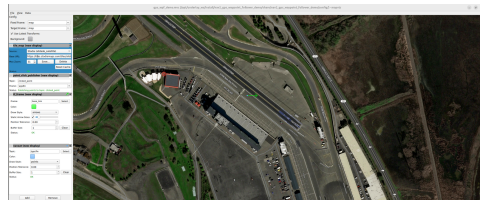
KPI (rule source)	Target	Measured
Obstacle reaction time	≤ 250 ms	47 ms
Min. average speed (hold-up rule)	≥ 1 mph	1.7 mph
Waypoint arrival accuracy	≤ 1 m	<1 m
Obstacle detection range	≥ 2 m	15 m
Full-course completion	≤ 5 min	field run pending

Field test 2026-05-21: 7 recoveries, 0 collisions around a person in the path.

Localization: control in the drift-free frame

- Dual EKF at 30 Hz: local frame fuses IMU + wheel odometry + ZED yaw; global frame adds GPS
- GPS is **unaided** (2–5 m), fused **differentially** behind a 6σ outlier gate — **no RTK** (IGVC §1.2)
- Per-step GPS deltas, not absolute fixes, stop $\pm 2\text{--}5$ m noise from jolting the costmap
- Safety-critical control runs in the **drift-free odometry frame**

A deliberate design choice, not a workaround.



GPS waypoints in the map frame; control in the drift-free odom frame

Cyber security: three risks, defense in depth

We ran a NIST risk assessment; per IGVC §11 here are our top three vulnerabilities.

Vulnerability	What we do today	Production hardening
Onboard network (default DDS)	Wi-Fi limited to team laptops; judges' wireless E-stop cuts motor power	ROS2 DDS security (per-node certs, encryption) + VLAN / WPA3
Web-UI hijack (no login)	500ms watchdog zeros drive on lost input; manual override; E-stop	TLS client certs + per-session tokens; Foxglove bound to localhost
GPS spoofing	6σ EKF rejects implausible jumps; GPS used as per-step correction only	RAIM multi-band receiver; authenticated RTK; dead-reckoning cross-check

Defense in depth: two independent *hardware* stops — a physical E-stop button and the judges' wireless relay — plus a software interlock; no single software compromise can move the vehicle.